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Reading AI-Written Code: An Eye-Tracking Study of Code Reviewers

Abstract

We recorded the eye movements of 24 computer-science graduate students while they reviewed code changes written either by humans or by an AI coding assistant. Reviewers made correct decisions equally often for both sources but spent less time reading AI-generated code.

1 Method

Each participant reviewed 16 Python code changes, 8 written by humans and 8 by an AI assistant, without being told the source. We define accuracy as the percentage of accept/reject decisions that matched the ground-truth label. We measured review time and total fixation duration on changed lines.

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2 Results

Decision accuracy was 81% for human-written changes and 79% for AI-written changes, a difference that was not statistically significant. Reviewers spent less time reading AI-generated code (Figure 1). Mean review time was 6.2 minutes for human-written and 4.9 minutes for AI-written changes.

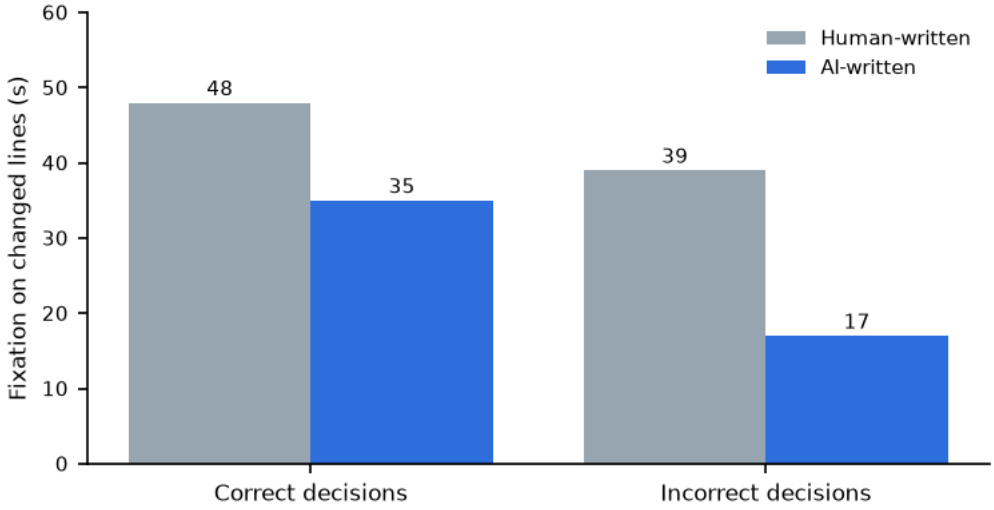


Figure 1: Total fixation duration (seconds) on changed lines, by code source and decision outcome.

3 Limitations

Participants were students, all code was Python, and code changes were short (under 60 lines).